SCOOT · Level 1 · Foundation

The traffic light that sees the queue before it forms

How SCOOT counts cars far up the road, predicts the queue at the lights, and nudges the timings a few seconds at a time.

6 ideas, each building on the last

1 What SCOOT is: the elastic plan

~192 small timing decisions per hour at one junction

2 Detectors and link profile units

1.96 vs 3.07 LPUs for a car and a bus at 13.5 m/s

3 Profiles and platoon dispersion

F = 0.263 smoothing factor on a 10-second link

4 The queue model and degree of saturation

90% SCOOT's usual degree-of-saturation target

5 Validation: making the model match the street

4 s clear-time error from a 12% SATO error

6 SCOOT against fixed time and actuation

12.8 s SCOOT mean delay in Lab F3 (fixed 19.0 s)

Three surprises to start

The detector is in the 'wrong' place on purpose.

Most actuated signals put their loops at the stop line. SCOOT puts them near the far end of the road, just after the previous junction. That gives it 10 to 20 seconds of warning: it knows a platoon is coming before any driver reaches the lights.

SCOOT never sees the queue.

There is no camera watching the queue. Every queue SCOOT uses is a prediction from its model. In our SUMO lab the model's queue still tracked the real queue closely, second by second.

It has been running since about 1980.

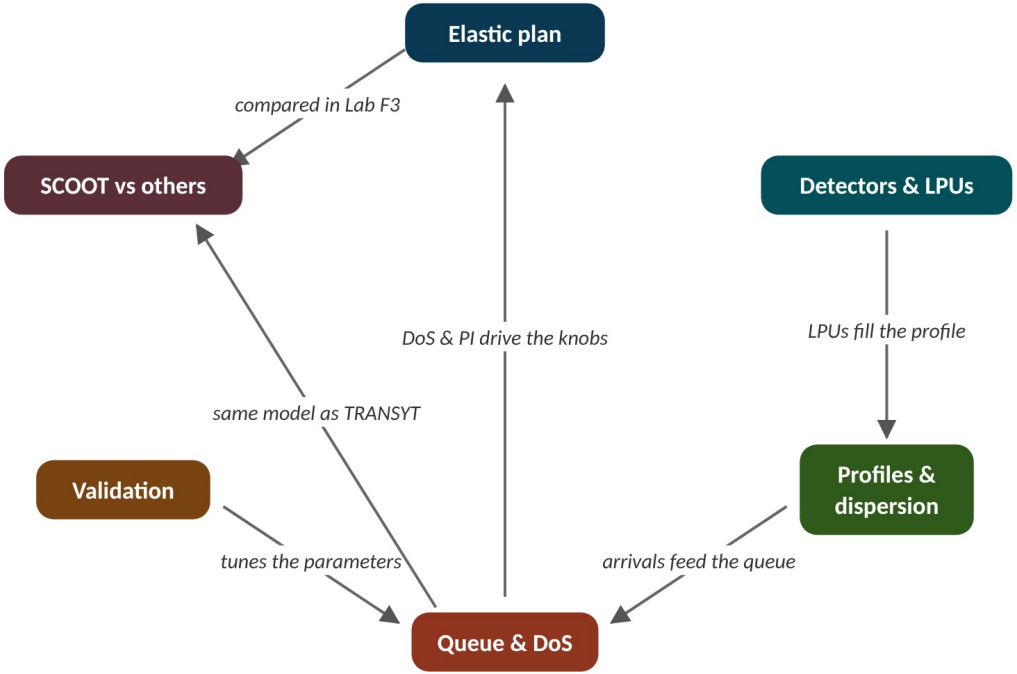
SCOOT was developed in the UK by the Transport and Road Research Laboratory (now TRL) and is used in hundreds of towns and cities. SCOOT 7 (2020) added connected-vehicle data and pedestrian features.

Words you will meet

Junction / node	A place where roads meet and traffic lights take turns. SCOOT calls each one a **node**.
Link	One approach road into a junction, from the detector to the stop line.
Stage	A set of movements that get green at the same time, for example 'main road both ways'.
Cycle	One full turn through all stages, for example 60 seconds.
Split	How the cycle is shared between stages: 'main road 35 s, side road 20 s'.

Offset	When one junction's cycle starts compared with its neighbour, so a green wave can form.
Platoon	A bunch of cars travelling together, usually released by the previous green.
Detector (loop)	A wire loop in the road that knows when a vehicle is over it.
Saturation flow	The fastest a queue can leave on green, roughly one car every two seconds per lane.

The map of today's ideas



1 What SCOOT is: the elastic plan

IN PLAIN WORDS

Think of a thermostat that nudges the heating a little every few minutes, rather than a timer that switches between 'winter' and 'summer' settings. Each nudge is small, so one wrong guess cannot do much harm, and the many nudges add up to follow the weather closely.

- Split, Cycle, Offset: three timing knobs
- Small, frequent changes: the elastic plan
- Central computer, a model of every link
- Three speeds: stage change, cycle, 5 minutes

~192

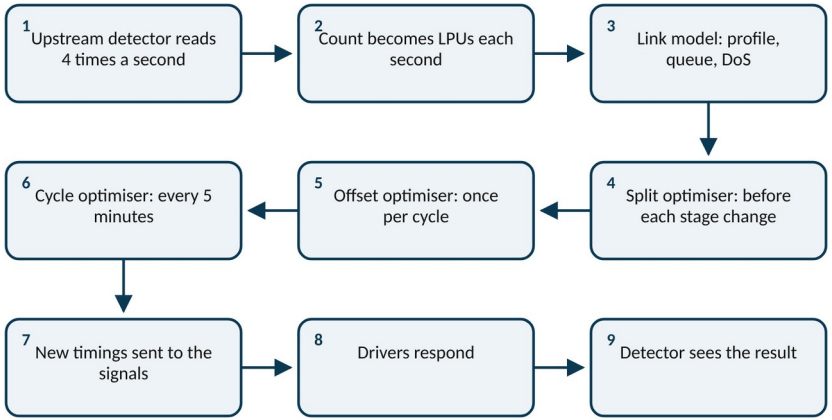
small timing decisions per hour at one junction

Worked example: How many decisions in one hour?

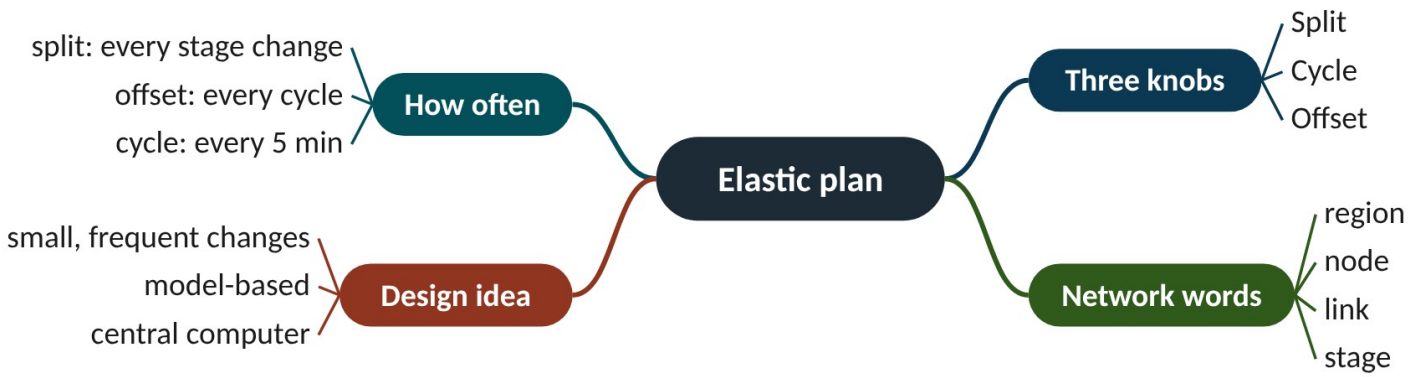
One junction, two stages, a 60 s cycle, in a region whose cycle is reviewed every 5 minutes.

1. Cycles per hour = 3600 ÷ 60 = **60**.
2. Split decisions: one before each stage change, two changes per cycle, so 60 × 2 = **120**.
3. Offset decisions: one per cycle, so **60**.
4. Cycle decisions: one per 5 minutes, so 60 ÷ 5 = **12**.

About 192 small decisions an hour at one junction. That is why each one can be small: there are plenty more to come.



Remember it: Elastic plan



PICTURE THIS · STOP 1: TOWN GATE

At the town gate a giant rubber band is stretched around the traffic light. Three dials on the pole are labelled S, C and O. A tiny engineer twangs the band every few seconds; it stretches, never snaps.

TRAP

SCOOT does not pick a new plan from a library of plans (that is plan selection, an older idea). It edits the plan it is already running, a few seconds at a time.

2 Detectors and link profile units

IN PLAIN WORDS

Imagine a doorman who charges every person one coin to enter, plus two coins for every second they block the door. A quick walker pays a little over one coin; someone pushing a wide trolley pays three. The bill measures how much of the doorway each person used, not just how many came in.

- Loops at the upstream end of each link
- Gives 10-20 seconds of warning
- LPU = vehicles + occupancy
- Long or slow vehicles count more

1.96 vs 3.07

LPUs for a car and a bus at 13.5 m/s

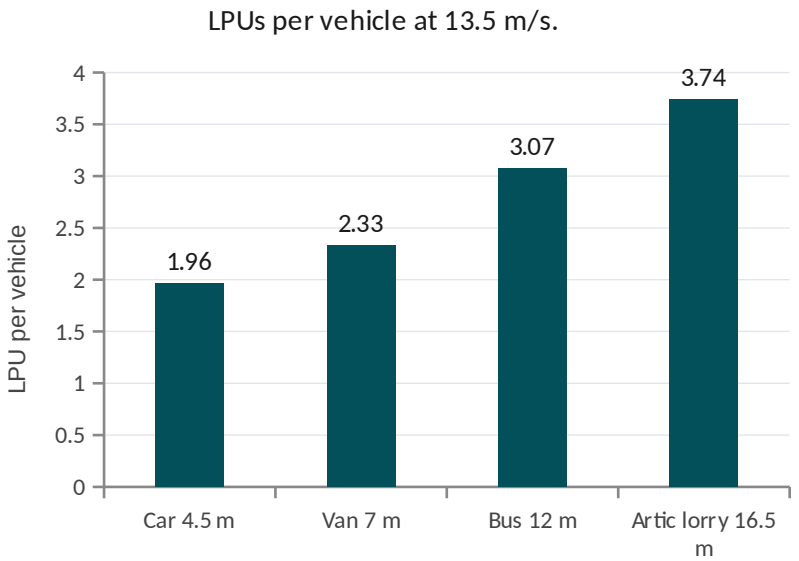
Worked example: A car and a bus at 13.5 m/s over a 2 m loop

occupancy time = (vehicle length + loop length) ÷ speed
LPU = 1 × vehicles + 2 × seconds occupied

Car length 4.5 m. Bus length 12 m. Speed 13.5 m/s (about 49 km/h).

1. Car occupancy: (4.5 + 2) ÷ 13.5 = 6.5 ÷ 13.5 = **0.48 s**.
2. Car LPU: 1 + 2 × 0.48 = **1.96 LPU**.
3. Bus occupancy: (12 + 2) ÷ 13.5 = 14 ÷ 13.5 = **1.04 s**.
4. Bus LPU: 1 + 2 × 1.04 = **3.07 LPU**.
5. Ratio: 3.07 ÷ 1.96 = **1.57**.

The bus counts as about one and a half cars, without the loop ever 'knowing' it was a bus. The SUMO validation lab measured 1.95 LPU per vehicle on average, very close to our car.



Remember it: Detectors & LPUs



PICTURE THIS · STOP 2: TOLL DOORMAT

Just past the gate, a doormat across the whole road is a cash register. Every vehicle pays ONE gold coin to roll on, then TWO coins for every second it sits there. A long red bus groans, paying three coins and dropping one on its foot.

TRAP

Do not think LPUs are 'vehicles'. A queue of 10 cars is about 20 LPUs in this lab. Always check whether a number is in vehicles or LPUs before you compare it.

3 Profiles and platoon dispersion

IN PLAIN WORDS

Picture a school class leaving the gate together and walking to the bus stop. The fast walkers pull ahead, the slow ones fall behind. At the bus stop the class arrives over a longer time than it took to leave, and the busiest moment is less busy.

- Cyclic flow profile: flow by second of the cycle
- Shifted by about 0.8 × cruise time
- Spread by Robertson's model
- Same total, flatter and later peak

F = 0.263

smoothing factor on a 10-second link

Worked example: A 10-second platoon on a 10-second link

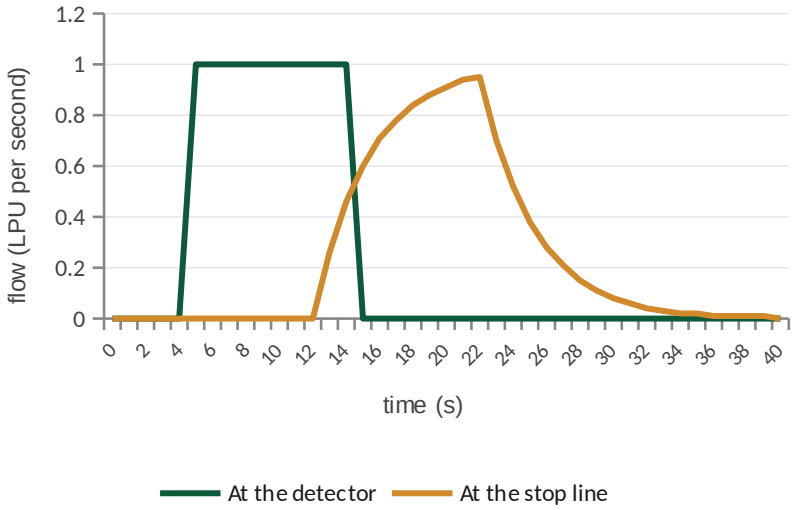
$$q_{stop}(t + \beta T) = F \cdot q_{det}(t) + (1 - F) \cdot q_{stop}(t + \beta T - 1)$$
$$F = 1 \div (1 + \alpha\, \beta\, T), \quad \alpha = 0.35,\ \beta = 0.8$$

A platoon crosses the detector at 1 LPU per second for 10 seconds. Cruise time T = 10 s.

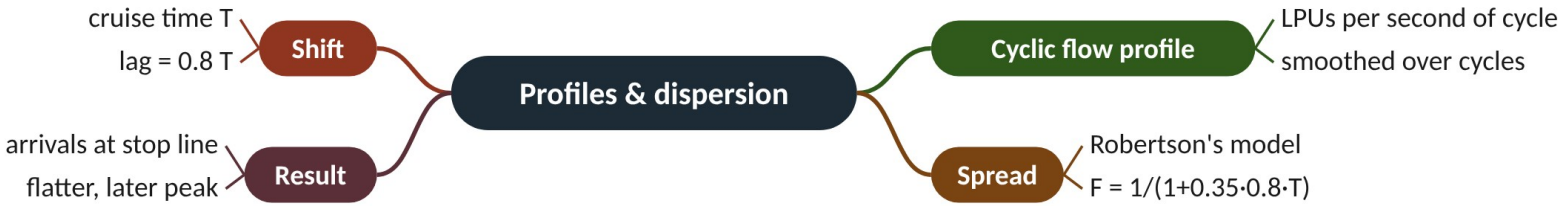
1. Lag βT = 0.8 × 10 = **8 s**: the first arrivals appear 8 s after the first detection.
2. F = 1 ÷ (1 + 0.35 × 0.8 × 10) = 1 ÷ 3.8 = **0.263**.
3. Second 1 at the stop line: 0.263 × 1 + 0.737 × 0 = **0.263**.
4. Second 2: 0.263 × 1 + 0.737 × 0.263 = **0.457**.
5. Second 3: 0.263 × 1 + 0.737 × 0.457 = **0.600**.
6. Keep going: the arrivals reach a peak of 0.95 after 10 s and then fade away over roughly 15 more seconds.

The same 10 LPUs arrive, but spread over about 25 s with a lower peak (0.95 instead of 1.0). Nothing is lost; it is only rearranged in time.

The same platoon at the detector and at the stop line.



Remember it: Profiles & dispersion



PICTURE THIS · STOP 3: MARCHING-BAND BRIDGE

On a bridge, a marching band of 35 players leaves in a tight block. A sign says 'walk at 80% speed'. By the far end of the bridge the trombones are way ahead and the drummers trail far behind, still playing.

TRAP

The lag is not the full cruise time. In Robertson's model the platoon's front arrives at $\beta T = 0.8\ T$, because the fastest drivers get there early. The rest arrive later in the tail.

4 The queue model and degree of saturation

IN PLAIN WORDS

A queue is a bathtub. Arrivals are the tap, always running. During red the plug is in, so the water rises. During green the plug is out and the water drains at a fixed speed (the saturation occupancy). If the tap is slower than the drain, the tub empties during green.

- Red fills the queue, green drains it at SATO
- Delay is the area under the queue
- DoS = demand ÷ what green can serve
- Aim for 90% or less; PI = delay + K·stops

90%

SCOOT's usual degree-of-saturation target

Worked example: One 60 s cycle on one link

DoS = (LPU arriving per cycle + residual queue) ÷ (effective green × SATO)
PI = delay + K × stops

Arrivals 0.2 LPU/s, steady. Effective red 35 s, effective green 25 s. SATO 0.6 LPU/s. No queue at the start.

1. Queue at the end of red: 0.2 × 35 = **7 LPU**.

2. In green the queue shrinks at 0.6 − 0.2 = 0.4 LPU/s, so it clears after 7 ÷ 0.4 = **17.5 s**.

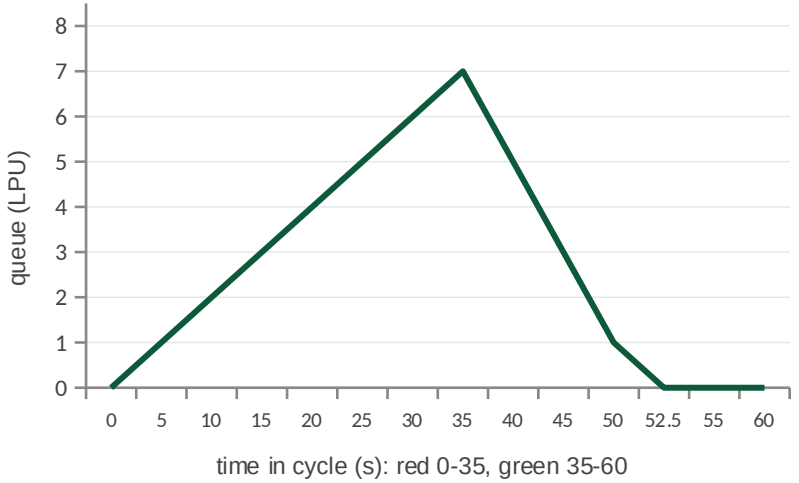
3. Delay = triangle areas: ½ × 35 × 7 + ½ × 17.5 × 7 = 122.5 + 61.25 = **183.75 LPU·s** per cycle.

4. Stops = everyone who arrived before the queue cleared: 0.2 × (35 + 17.5) = **10.5 LPU**.

5. Demand per cycle = 0.2 × 60 = 12 LPU. Capacity = 25 × 0.6 = 15 LPU. DoS = 12 ÷ 15 = **0.80**.

DoS is 80%, under the 90% target, so this link has some spare green to give away.

The modelled queue over one cycle.



Remember it: Queue & DoS



PICTURE THIS · STOP 4: BATHTUB AT THE STOP LINE

At the stop line stands a huge bathtub with a tap that never stops. When the light is red, a rubber plug shaped like a red light is in. On green the plug pops and water swirls out. A gauge on the side is painted red above the 90% mark.

TRAP

A DoS of 100% does not mean the road is jammed forever; it means the green is exactly used up. Above 100% a residual queue carries into the next cycle and grows. Also: DoS uses effective green, not the displayed green.

5 Validation: making the model match the street

IN PLAIN WORDS

Validation is like tuning a guitar by ear. The model is the guitar; the street is the tuning fork. You pluck, listen, and turn the peg until they agree. A perfectly built guitar still plays wrong notes if nobody tunes it.

- The model is only as good as its parameters
- Engineer compares model with the street
- Queue clears early in model → SATO too high
- Lab: SATO 0.67 / 0.89, lags 3 s / 4 s

4 s

clear-time error from a 12% SATO error

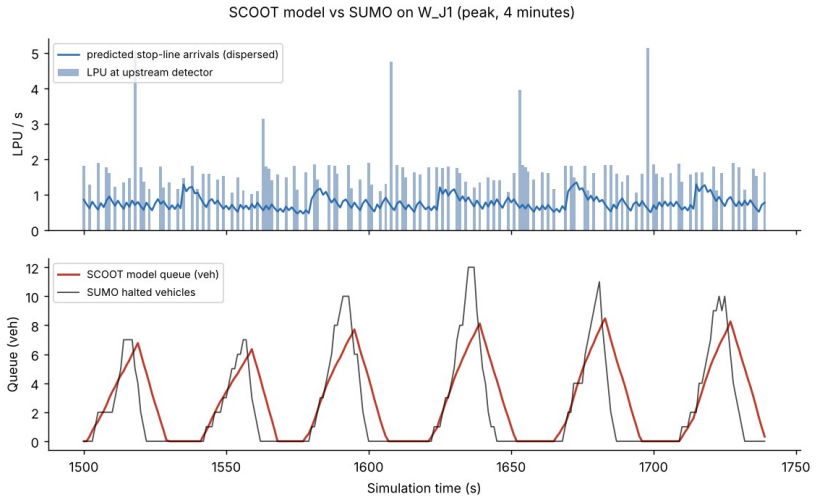
Worked example: Fixing SATO from a stopwatch

queue clear time = start lag + queue ÷ SATO
SATO = queue ÷ (observed clear time − start lag)

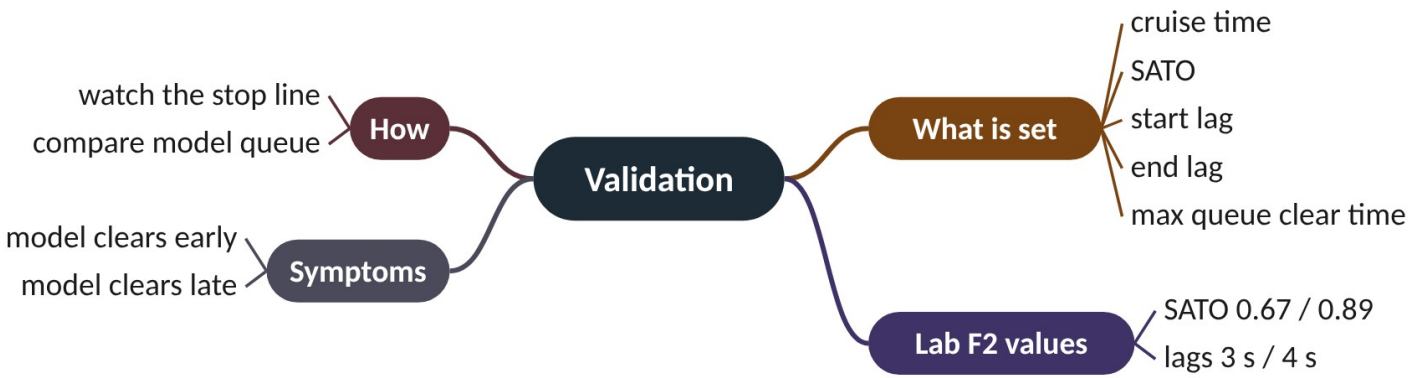
A queue of 10 cars (about 19.5 LPU) is waiting. Model SATO 0.67 LPU/s, start lag 3 s. The engineer sees the queue clear 36 s after green starts.

1. Model clear time: 3 + 19.5 ÷ 0.67 = 3 + 29.1 = **32.1 s**.
2. Observed: **36 s**. The model clears about 4 s early, so its SATO is too high.
3. Corrected SATO = 19.5 ÷ (36 − 3) = 19.5 ÷ 33 = **0.59 LPU/s**.
4. Check: 3 + 19.5 ÷ 0.59 = 3 + 33.0 = **36.0 s**. It now matches.

A 12% change in SATO moved the model's clear time by 4 s. Small parameter errors give big errors in delay and DoS.



Remember it: Validation



PICTURE THIS · STOP 5: GUITAR SHOP

Beside the stop line a guitar shop has a giant guitar strung across the road. An engineer with a stopwatch in her teeth twists a tuning peg marked SATO. The peg has two smaller pegs labelled 3 and 4.

TRAP

Do not 'fix' a model by changing the target DoS or the optimiser. If the queue picture is wrong, every decision built on it is wrong. Validate the model first.

6 SCOOT against fixed time and actuation

IN PLAIN WORDS

Fixed time is a printed bus timetable. Actuation is a driver who leaves as soon as nobody else is running for the bus. SCOOT is a dispatcher with a radio who hears that a crowd is coming in 15 seconds and plans for it.

- Fixed time: a stored plan (Webster, TRANSYT)
- Actuated: reacts to gaps at the stop line
- SCOOT: predicts from upstream, adapts gently
- Lab F3: 19.0 vs 14.5 vs 12.8 s delay

12.8 s

SCOOT mean delay in Lab F3 (fixed 19.0 s)

Worked example: Webster for a two-stage junction

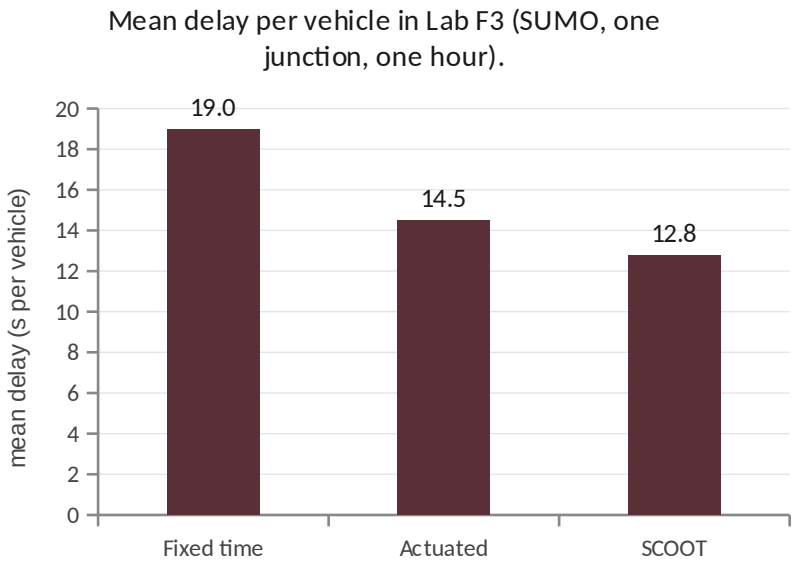
$$C_0 = (1.5\,L + 5) \div (1 - Y)$$

$$g_i = (C_0 - L) \times y_i \div Y$$

Lost time L = 10 s. Main road flow ratio y_A = 0.35, side road y_B = 0.25.

1. Y = 0.35 + 0.25 = **0.60**.
2. C_0 = (1.5 × 10 + 5) ÷ (1 − 0.60) = 20 ÷ 0.40 = **50 s**.
3. Effective green to share = 50 − 10 = **40 s**.
4. Main road: 40 × 0.35 ÷ 0.60 = **23.3 s**. Side road: 40 × 0.25 ÷ 0.60 = **16.7 s**.

A good fixed plan for these flows: 50 s cycle, 23.3 s and 16.7 s of effective green. It stays good only while the flows stay the same, which is why SCOOT adapts.



Remember it: SCOOT vs others



PICTURE THIS · STOP 6: FINISH LINE AT THE CONTROL CENTRE

Outside the control centre three runners finish a race. A robot with a printed timetable on its chest crosses at 19, a jumpy rabbit at 14 and a half, and a calm dispatcher wearing headphones at just under 13. On the podium sits a set of scales labelled '1.5 L + 5'.

TRAP

Webster's formula blows up as Y approaches 1: at Y = 0.9 and L = 10, C0 = 200 s. Real controllers cap the cycle, and when Y ≥ 1 the junction is simply over capacity: no cycle length can fix it.

Three ways to run a traffic light

	Fixed time	Actuated	SCOOT
Detectors	none	at the stop line	upstream, 10-20 s before the stop line
Decides with	a stored plan	gaps between vehicles	a model of every link
Coordinates junctions?	yes, if designed so	no	yes, through offsets and a shared cycle
Reacts to change	no	every second	small steps: stage change, cycle, 5 min
Lab F3 delay	18.96 s	14.48 s	12.79 s

Things beginners mix up

This…	…is not this	The fact that tells them apart
Vehicles	**LPUs**	LPUs add occupancy, so a car is about 2 LPU and a bus about 3.
Cruise time T	**Lag βT**	The platoon's front arrives at 0.8 T, not T.
Displayed green	**Effective green**	Effective green starts a start lag late and runs an end lag into amber.
DoS 100%	**Gridlock**	100% means the green is just used up; queues grow only above it.
Elastic plan	**Plan selection**	SCOOT edits the running plan; plan selection swaps whole stored plans.

Check yourself (1 of 2)

A 7 m van crosses a 2 m loop at 13.5 m/s. Using this lab's LPU rule, how many LPUs?

A. 1.00

B. 1.67

C. 2.33

D. 3.07

Arrivals 0.25 LPU/s, effective green 30 s, SATO 0.5 LPU/s, cycle 60 s, no residual queue. DoS?

A. 0.50

B. 0.83

C. 1.00

D. 1.20

Answers are in the speaker notes and in the study guide's answer key.

Check yourself (2 of 2)

In Lab F3 which controller had the lowest mean delay?

A. Fixed time

B. Actuated

C. SCOOT

D. All the same

Webster with Y = 0.95 and L = 10 gives C_0 = 400 s. What is the right conclusion?

A. Use a 400 s cycle

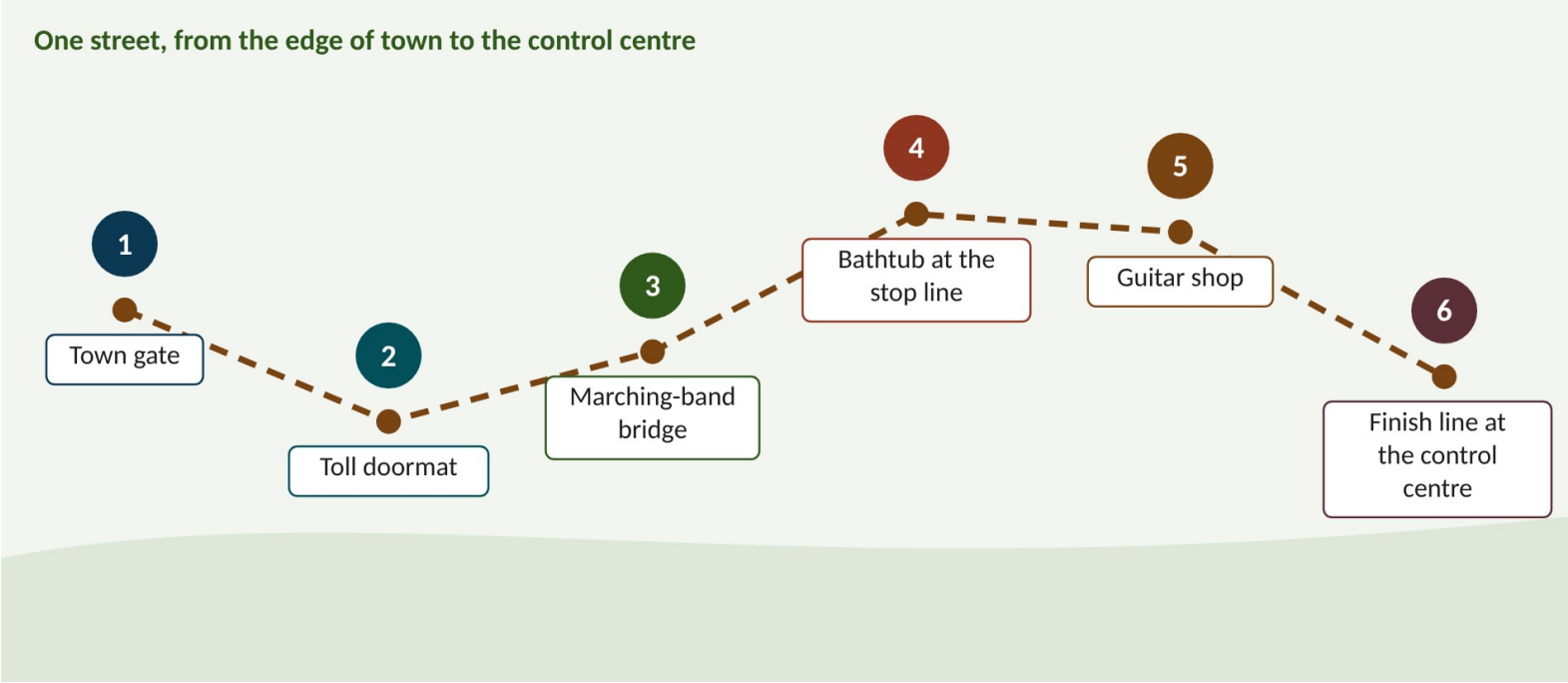
B. The junction is near capacity; cap the cycle and look at capacity, not timing

C. Webster is wrong for all junctions

D. Set Y to 0.6

Answers are in the speaker notes and in the study guide's answer key.

Walk the memory route



1 Town gate = Elastic plan · 2 Toll doormat = Detectors & LPUs · 3 Marching-band bridge = Profiles & dispersion · 4 Bathtub at the stop line = Queue & DoS · 5 Guitar shop = Validation · 6 Finish line at the control centre = SCOOT vs others

Numbers to remember

- Loop read 4 times a second; 10-20 s warning
- Car ≈ 1.96 LPU, bus ≈ 3.07 LPU (13.5 m/s, 2 m loop)
- $\alpha = 0.35$, $\beta = 0.8$; T = 10 → F = 0.263
- DoS target ≈ 90%
- Lab F2: SATO 0.67 / 0.89 LPU/s per lane, lags 3 / 4 s
- Lab F3 delay: 18.96 / 14.48 / 12.79 s
- Webster: L 10, Y 0.6 → C_0 = 50 s

Keep going

Study guide: SCOOT_Foundation_Study_Guide.pdf · Interactive labs and SUMO code: smrahmanbd.github.io/scoot-learning-lab